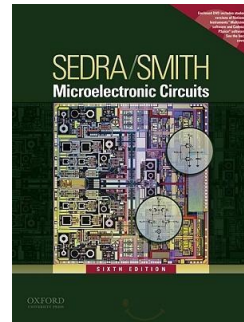


Partie 1 : Dispositifs et Circuits de base

- 3 : Diodes



Chapter 3



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ELEC1530 – Partie 1 : Diodes - 1

1

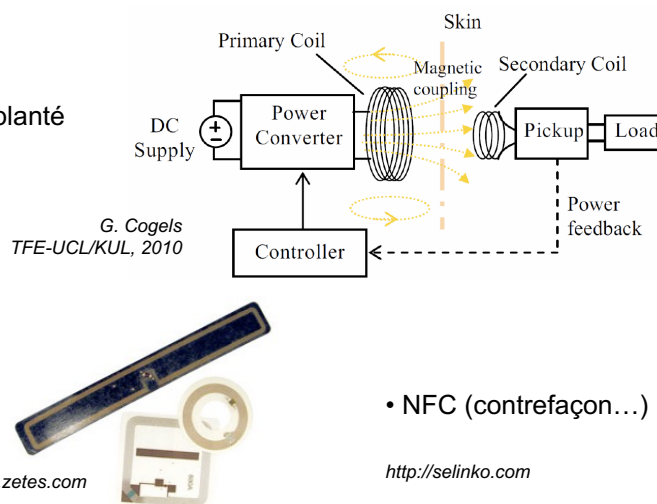
Introduction : Systèmes de transfert de puissance et données par couplage inductif

Exemples :

- Biomédical implanté (pacemakers...)

- Etiquette RFID (logistique, passeports...)

- NFC (contrefaçon...)

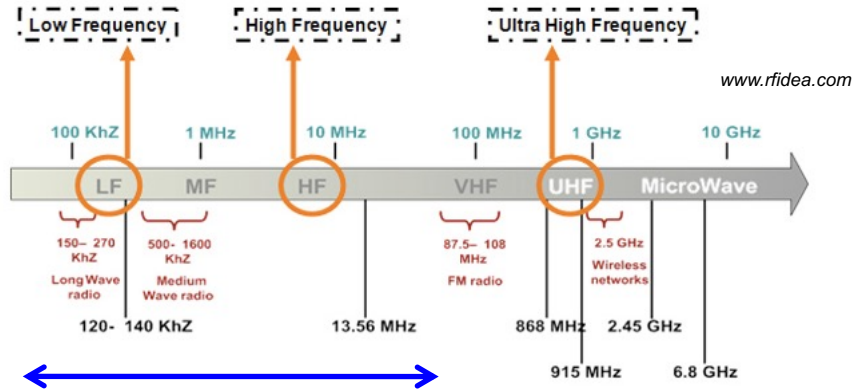


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2

Fréquence de la porteuse



Couplage inductif avec bobines

- Distances : mm ... m

- Tension ac sur bobine secondaire : ... 100 mV → 10 V ...

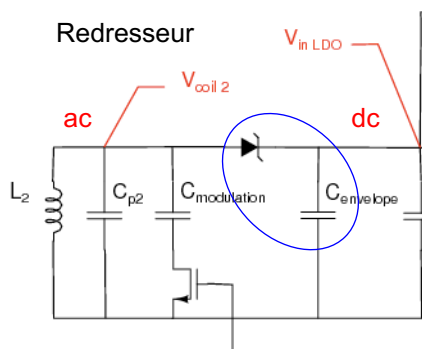


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3

Tension ac sur bobine → Tension dc pour circuits (1,5 ... 5 V)

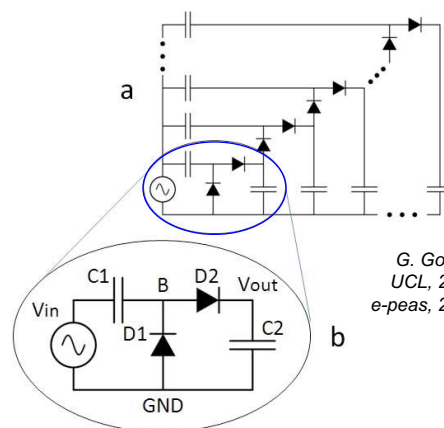


G. Cogels
TFE-UCL/KUL, 2010

⇒ **DIODES !**

Elévateur

$V_{out} \gg V_{in}$



G. Gosset
UCL, 2009
e-peas, 2017



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Plan du cours

- 4.1 Diode idéale
- 4.2 Diode réelle
- 4.3 Modèles (DC, AC)
- 4.4 Diodes Zener → LELEC1755, 2660...
- 4.5 Circuits redresseurs → LELEC2532, 2660 ...
- 4.6 Diodes spéciales (LED, PV) → LELEC2330 ...

4.1 Diode idéale

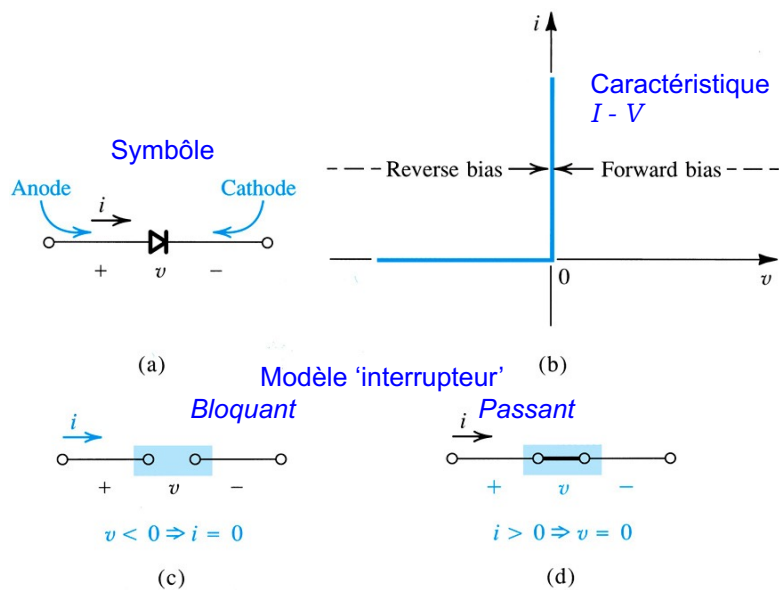
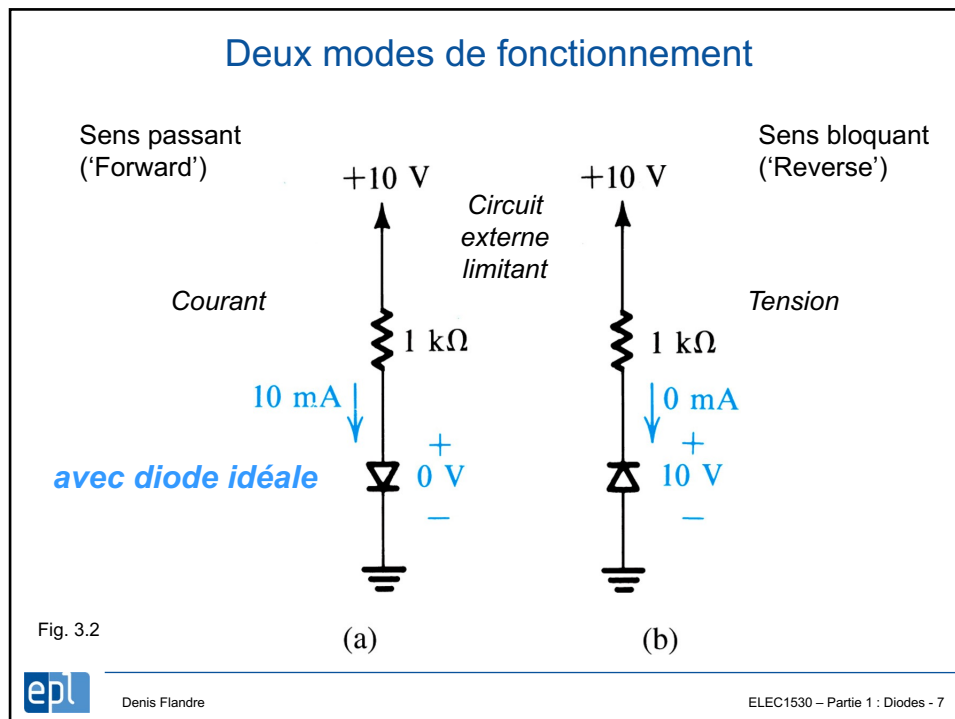
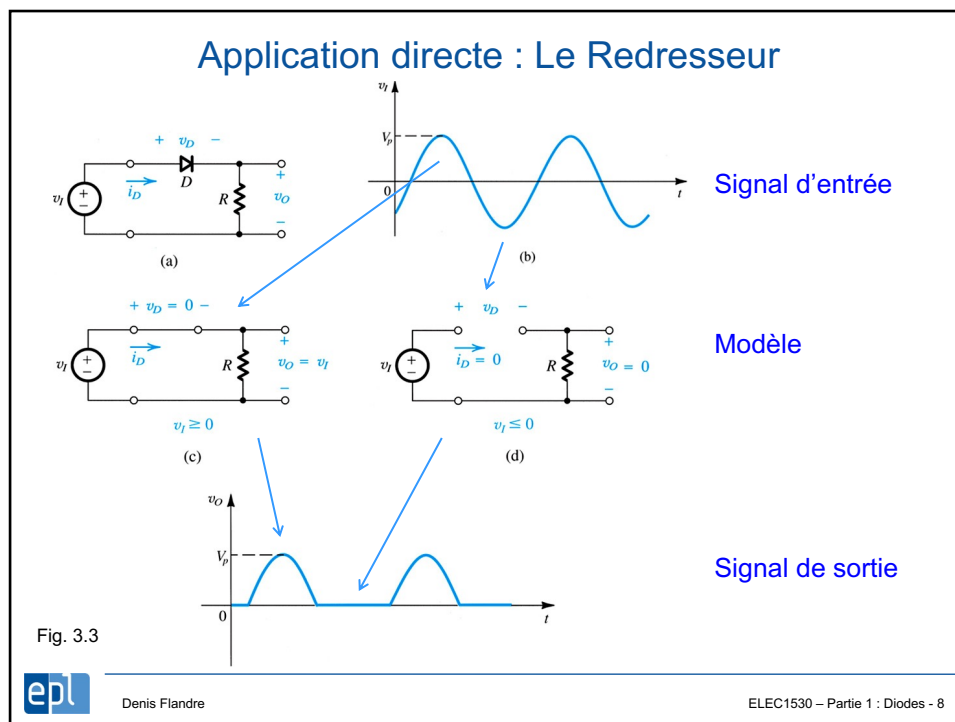


Fig. 3.1

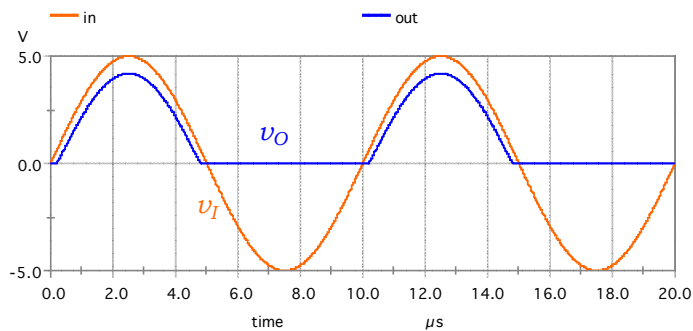
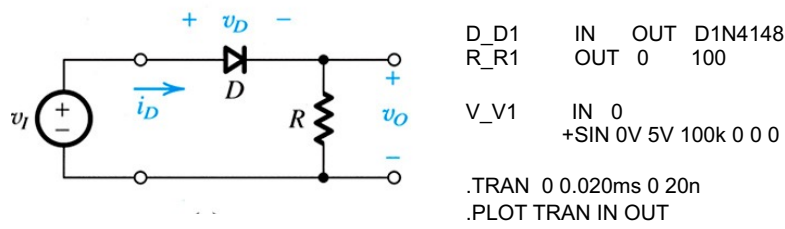


7



8

Simulation SPICE avec diode de la Sedra_Lib

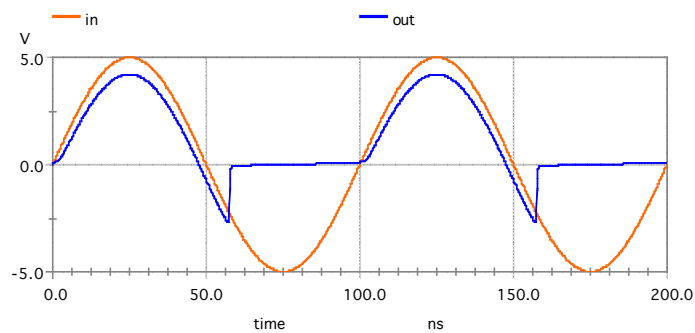


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@ 10 MHz



* Model for 1N4148 Diode (from EVAL library in PSpice)

```

.model D1N4148 D(Is=2.682n N=1.836 Rs=.5664 Ikf=44.17m Xti=3 Eg=1.11 Cjo=4p
+ M=.3333 Vj=.5 Fc=.5 Isr=1.565n Nr=2 Bv=100 Ibv=100u Tt=11.54n)
    
```

⇒ *Modèles DC et AC plus complets !*



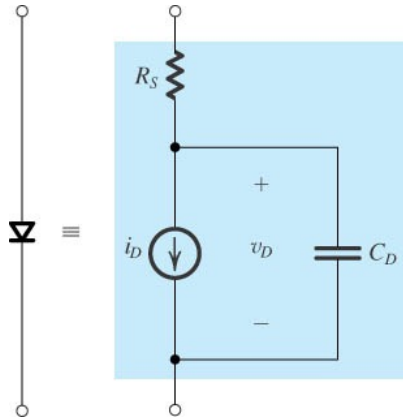
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Modèle SPICE

* Model for 1N4148 Diode (from EVAL library in PSpice)
 .model D1N4148 D(Is=2.682n N=1.836 Rs=.5664 Ikf=44.17m Xti=3 Eg=1.11 Cjo=4p
 + M=.3333 Vj=.5 Fc=.5 Isr=1.565n Nr=2 Bv=100 Ibv=100u Tt=11.54n)



$$i_D = I_S (e^{v_D/nV_T} - 1)$$

$$C_D = C_d + C_j = \frac{\tau_T}{V_T} I_S e^{v_D/nV_T} + C_{j0} \left(1 - \frac{v_D}{V_0}\right)^m$$

(LELEC1755)



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4.2 Caractéristiques I-V des diodes

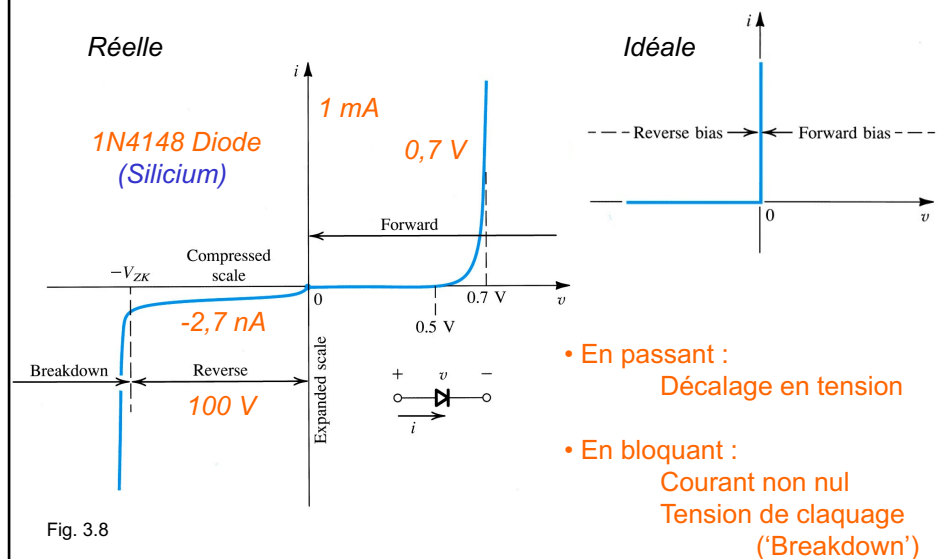


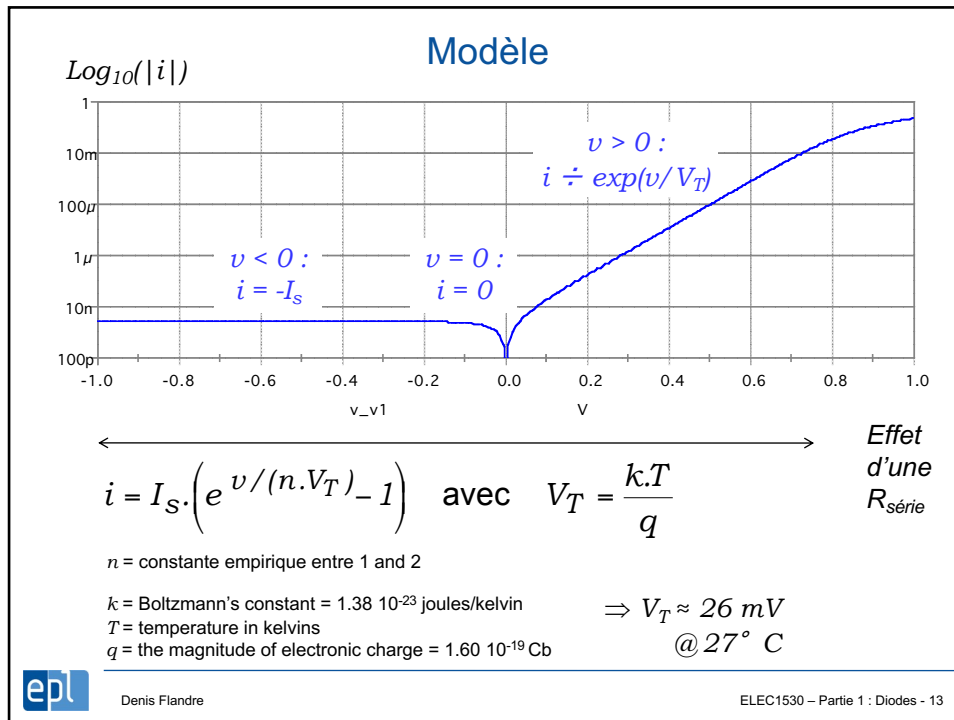
Fig. 3.8



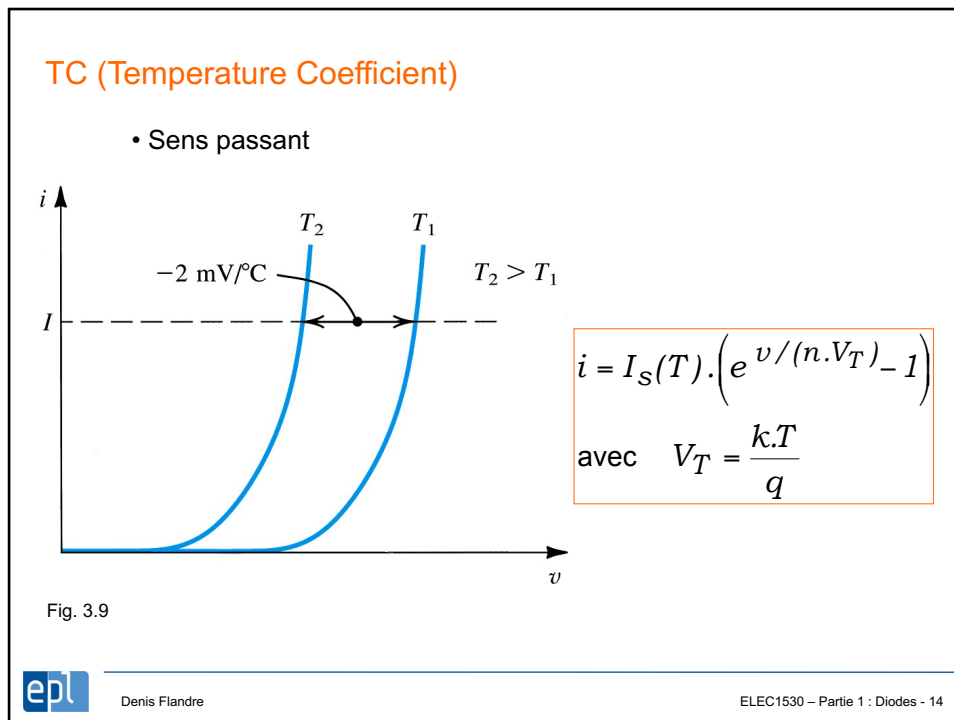
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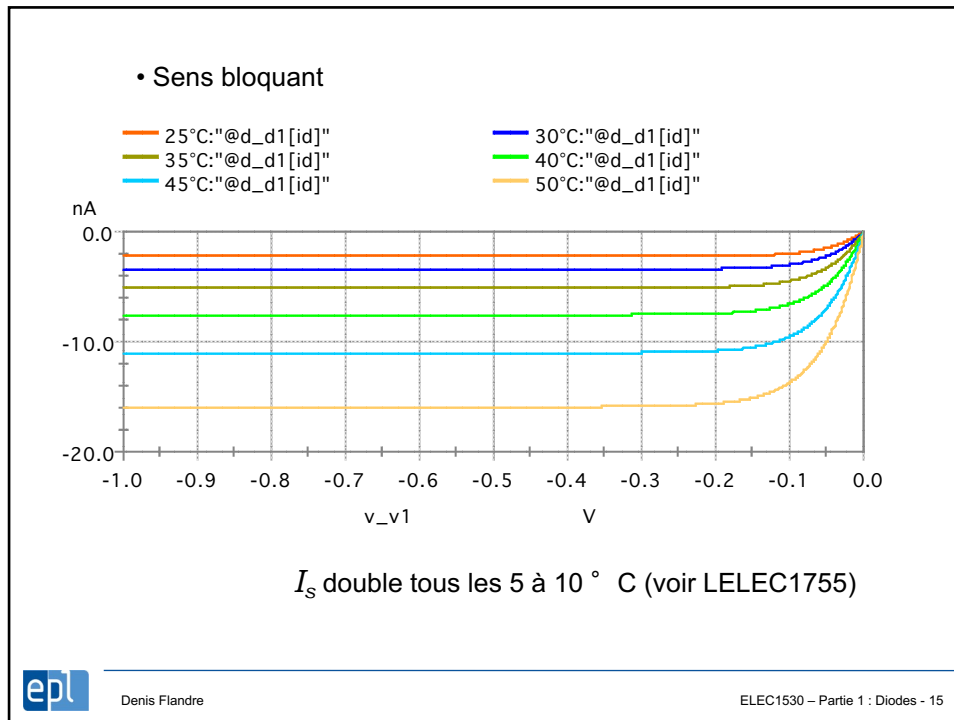
12



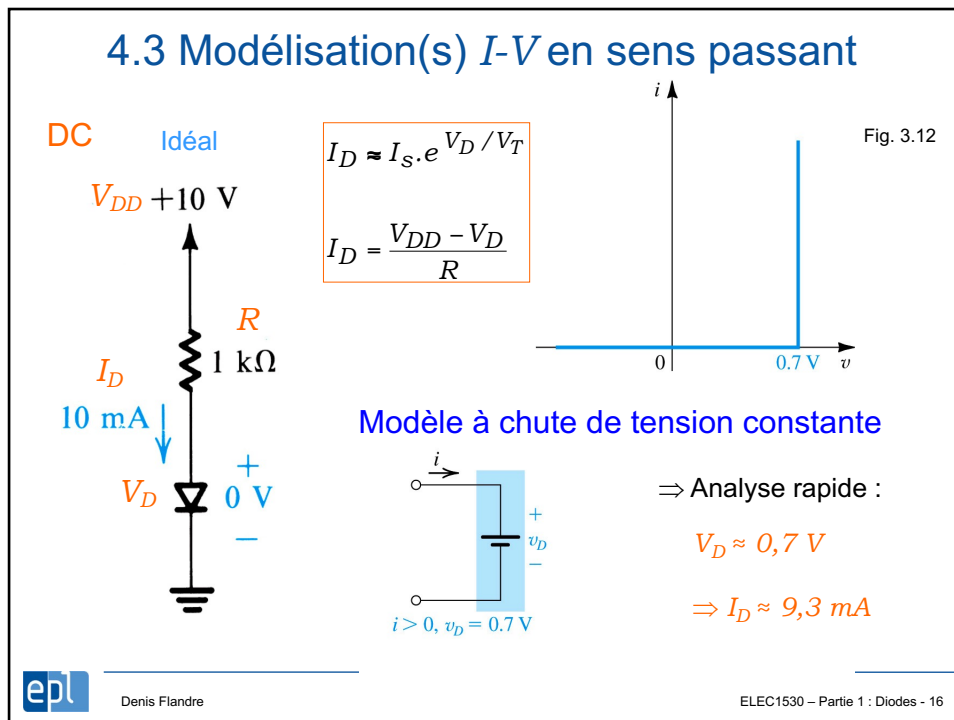
13



14

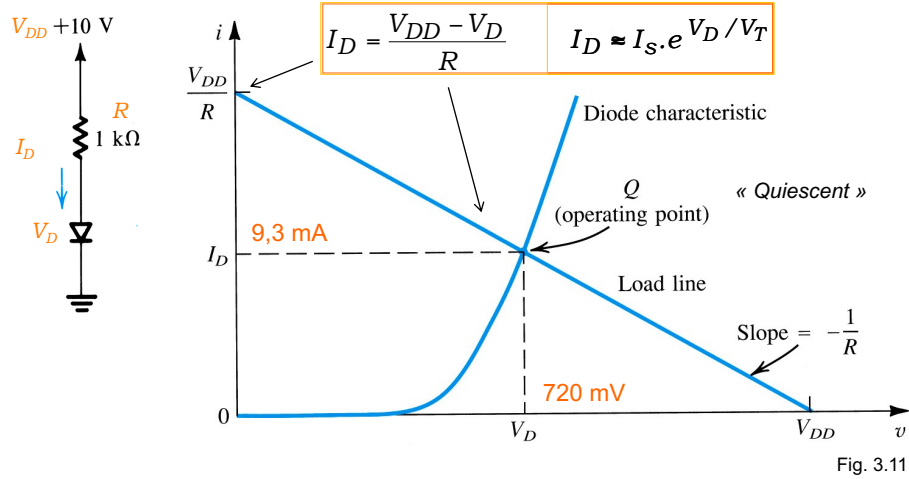


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Modèle exponentiel - Analyse graphique : Diagramme de charge



SPICE : .OP (VDD=10V, R=1kΩ, diode ...)

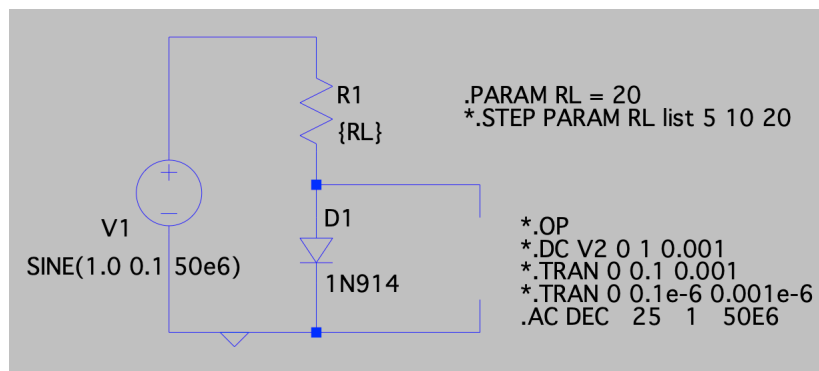


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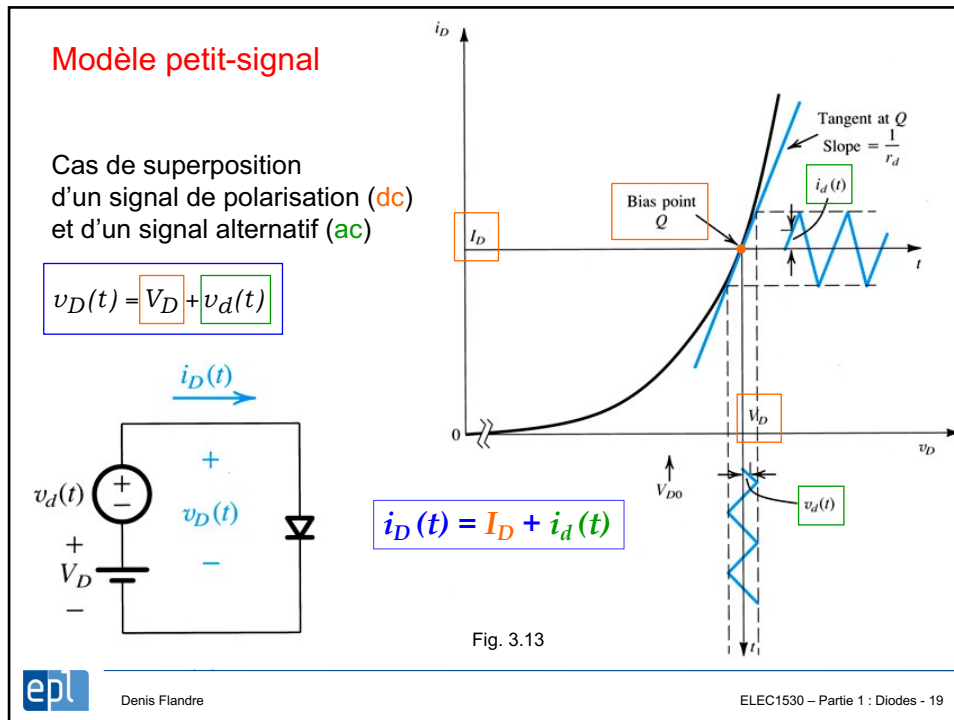
SPICE Demo



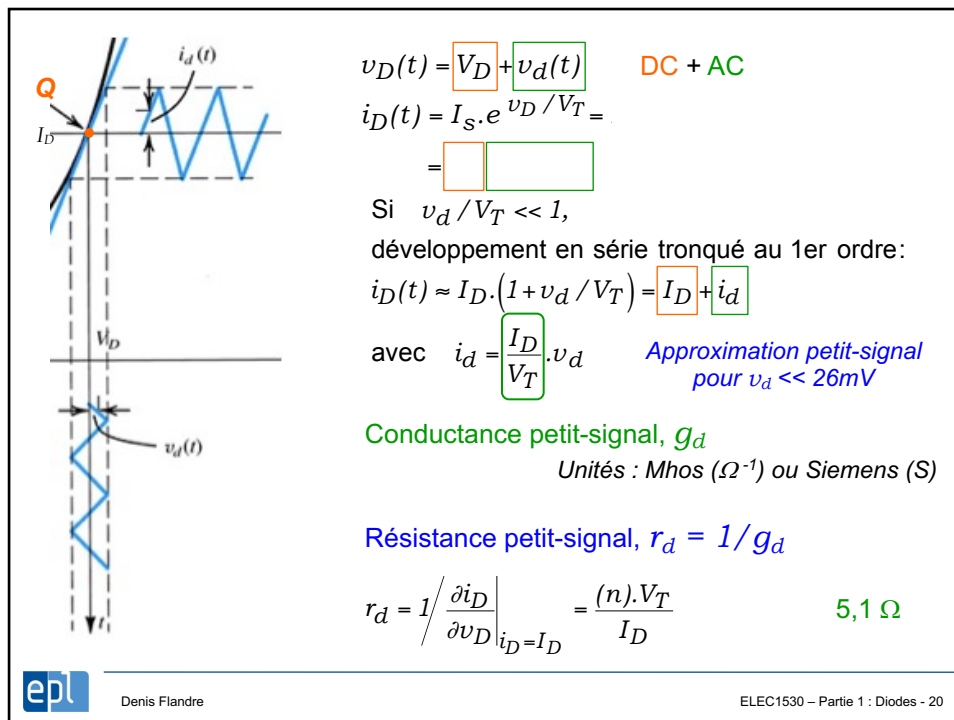
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Utilisation en régulation de tension

$$V^+ = V_{DD} (dc) + v_s (ac)$$

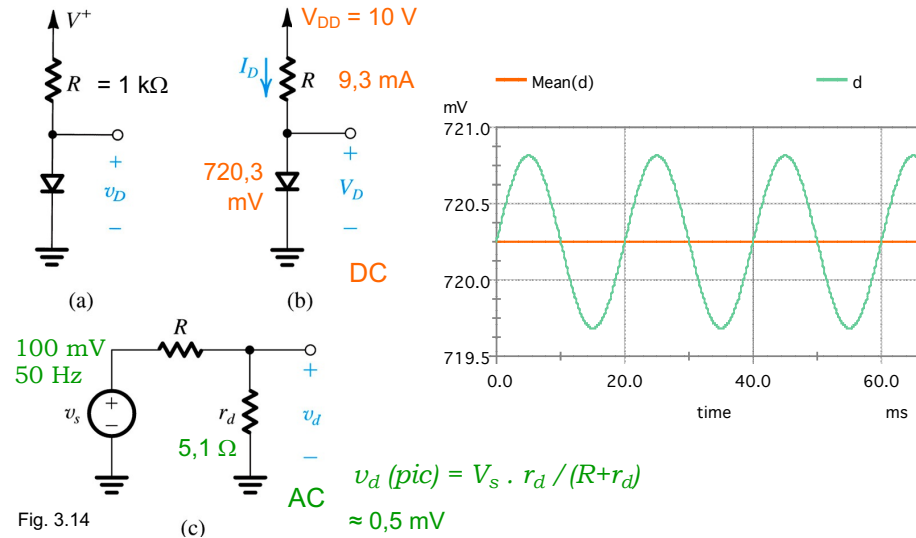


Fig. 3.14

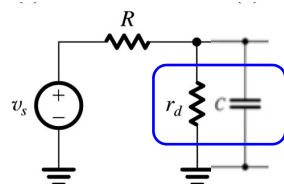
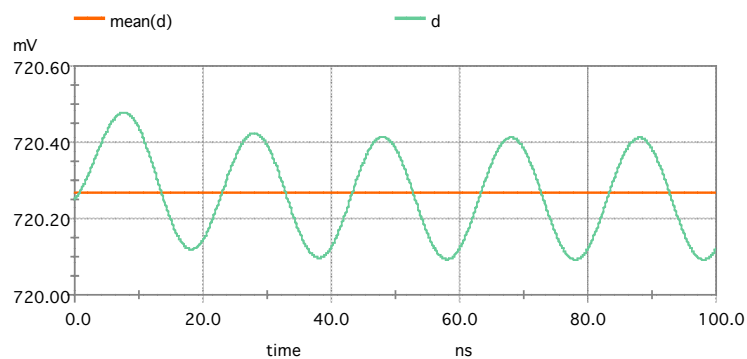


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$$V_s = 100 \text{ mV @ } 50 \text{ MHz}$$



Modèle petit-signal (HF) de la diode

$$C = 2.263 \text{ nF}, f_c = 1/(2\pi RC) \approx 13.7 \text{ MHz}$$



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ELEC1530 – Partie 1 : Diodes - 22

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Contenu du chapitre

Méthodologie

- 4.1 Diode idéale
 - Caractéristique I-V 1. Mesure / Simulation SPICE
 - Redresseur
- 4.2 Diode réelle
 - Régions passante / bloquante, Effet de la température
- 4.3 Modèles 2. Point de fonctionnement (DC Bias)
 - Seuil, Exponentiel, Graphique, Diagramme de charge, Petit signal
 - Régulation de tension 3. Réponse du circuit aux signaux
 - 4.4 Diodes Zener : claquage, spécifications
- 4.5 Circuits redresseurs 4. Simulations, performances
 - Half-wave, Peak
- 4.6 Diodes spéciales